Row Reduction and Row Echelon Forms

Recall from last week:

- A **linear equation** in the variable $x_1, \dots, x_n$ is an equation that can be put in the following form
$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$
where b and the coefficients $a_1, \dots, a_n$ are real or complex numbers. A **system of linear equations** is a collection of one or more linear equations involving the same variables. A **solution** to the linear system is a list of $(s_1, \dots, s_n)$ of numbers that are solutions to each of the linear equations in the system.
- A linear system is **consistent** if there is a solution and **inconsistent** if there is no solution.
- A linear system has either no solution, exactly one solution, or infinitely many solutions.
- The following elementary row operations preserve solutions.
  1. Replace one row by the sum of itself and a multiple of another row.
  2. Interchange two rows.
  3. Multiply all entries in a row by a non-zero constant.
- Two matrices are called **row equivalent** if there is a sequence of elementary row operations changing one matrix to another.
- If the augmented matrices of two linear systems are row equivalent, then they both have the same solution set.
- Proof Techniques
  1. Quantifiers and counterexamples.
  2. Direct Proofs.
  3. Proof by cases.
  4. Proof by contrapositive.

Definition 1. *A matrix is in* **row echelon form** *if:*

1. *All rows consisting of only zeros are at the bottom.*
2. *The leading coefficient (also called the pivot) of a non-zero row is always strictly to the right of the leading coefficient of the row above it.*

1.

(a) Give an example of a matrix in echelon form.

(b) Give an example of a matrix not in echelon form.

Definition 2. *A matrix in row echelon form is in* **reduced row echelon form** *if*

(a) The leading entries are all 1.

(b) Each leading 1 is the only non-zero entry in its column.

(c) Give an example in reduced row echelon form.

Fact: Each matrix is row equivalent to one and only one reduced echelon matrix.

Definition 3. *A* **pivot position** *in a matrix A is a location in A that corresponds to a leading 1 in the reduced row echelon form. A* **pivot column** *is a column that has a pivot position.*

Row Reduction Algorithm (Gaussian Elimination).

(a) The leftmost non-zero column is a pivot column.

(b) Select a non-zero entry in the pivot column as pivot. Interchange rows to move this entry to the top.

(c) Use elementary row operations to create zeros below the pivot.

(d) Apply (a)-(c) to the submatrix obtained by removing pivot row and column. Repeat until the matrix is in row echelon form.

(e) Beginning with rightmost pivot and working upwards and to the left, create 0's above each pivot. Scale pivot positions entries to be 1.

2. Row-reduce the following matrix:

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$$

3. Use row-reduction to solve the linear system

$$\begin{cases} x_1 + 3x_2 + 4x_3 = 7 \\ 3x_1 + 9x_2 + 7x_3 = 6 \end{cases}$$

Definition 4. *The variables corresponding to the pivot columns in an augmented matrix are called* **basic** *and the others are called* **free**.

Facts: A linear system is consistent if and only if the rightmost column in the augmented matrix is not a pivot column. If the system is consistent, then it has a unique solution when there are no free variables and infinite solutions when there is one or more free variables.

4. Find the general solution of the augmented matrix :

$$\left(\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right)$$

Definition 5. *A matrix with 1 column is called a* **column vector***. A matrix with one row is called a* **row vector***.*

We can add vectors of the same size by adding the corresponding components; namely, if $\vec{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$, then $\vec{u} + \vec{v} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix}$. Similarly, we can scale a vector by a real number c, by multiplying all the components by c; namely, $c\vec{u} = \begin{bmatrix} cu_1 \\ \vdots \\ cu_n \end{bmatrix}$.

Definition 6. *Given $\vec{v}_1, \ldots, \vec{v}_p \in \mathbb{R}^n$ and scalars $c_1, \ldots c_p$, the vector*

$$\vec{y} = c_1\vec{v}_1 + \ldots c_p\vec{v}_p$$

is called the **linear combination** *of $\vec{v}_1, \ldots, \vec{v}_p$ with weights $c_1, \ldots, c_p$.*

5. Suppose $\vec{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $\vec{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$; and $\vec{w} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$. Can $\vec{w}$ be expressed as a linear combination of $\vec{u}$ and $\vec{v}$?

6. Suppose $\vec{a} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$; $\vec{b} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$; $\vec{c} = \begin{bmatrix} 5 \\ -6 \\ 8 \end{bmatrix}$; and $\vec{d} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$. Can $\vec{d}$ be expressed as a linear combination of $\vec{a}$, $\vec{b}$, and $\vec{c}$?

A vector equation

$$x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$$

has the same solutions as the augment matrix

$$[\vec{a}_1\ \vec{a}_2\ \ldots\ \vec{a}_n \big| \vec{b}].$$

Hence $\vec{b}$ is a linear combination of $\vec{a}_1, \ldots, \vec{a}_n$ if and only if there is a solution to this matrix.

Definition 7. *If $\vec{v}_1, \ldots, \vec{v}_p \in \mathbb{R}^n$, then the set of all linear combinations of $\vec{v}_1, \ldots, \vec{v}_p$ is called the* **span** *of $\vec{v}_1, \ldots, \vec{v}_p$. We write this set as Span$\{\vec{v}_1, \ldots, \vec{v}_p\}$.*

7.

(a) Suppose $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$; $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$; and $\vec{b} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$. Is $\vec{b}$ in Span$\{\vec{u}, \vec{v}\}$?

(b) Does there exist $\vec{b}$ not in Span$\{\vec{u}, \vec{v}\}$?